APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 7)

SCENARIO: RELATIONAL MERGING OF MARKETING & TRANSACTION TABLES

INSTRUCTIONS FOR LEARNER: Complete the following four tasks in your Jupyter Notebook during class. Follow the structured workflow: **Table Inspection -> Relational Merge -> Aggregation -> Executive Interpretation**. Do not peek at instructor solution keys until completed.

TASK 1: TABLE INSPECTION & KEY VERIFICATION

Liam Vance (Marketing Manager) provides two separate tables from the promotional trial:

```python
import pandas as pd

orders_table = {
    "order_id": ["ORD-501", "ORD-502", "ORD-503", "ORD-504", "ORD-505"],
    "client_id": ["CLI-10", "CLI-20", "CLI-30", "CLI-40", "CLI-50"],
    "spend_usd": [320.00, 75.00, 890.00, 210.00, 540.00]
}

promo_table = {
    "client_id": ["CLI-10", "CLI-30", "CLI-50", "CLI-60"],
    "promo_code": ["PROMO-A", "PROMO-B", "PROMO-A", "PROMO-C"]
}
```

1. Import `pandas` as `pd` and construct DataFrames `df_orders` and `df_promo`.
2. Inspect the columns of both tables and identify the common key column used for joining.

TASK 2: EXECUTING INNER VS LEFT JOINS

1. Perform an **Inner Join** (`how='inner'`) combining `df_orders` and `df_promo` on the common key column using `pd.merge()`. Print the resulting DataFrame and note its row count.
2. Perform a **Left Join** (`how='left'`) combining `df_orders` (left) and `df_promo` (right). Print the resulting DataFrame and explain why `CLI-20` and `CLI-40` appear with `NaN` in the `promo_code` column.

TASK 3: FILLING MISSING VALUES & GROUPING AGGREGATION

Using your Left-Joined DataFrame from Task 2:

1. Fill missing `promo_code` values with the string `"No Promo"` using `.fillna()`.
2. Use `.groupby()` grouped by `promo_code` to calculate:
    - Total spend (`spend_usd`: `sum`)
    - Average spend (`spend_usd`: `mean`)
    - Order count (`order_id`: `count`)
3. Print the aggregated campaign summary table.

TASK 4: EXECUTIVE MARKETING MEMO

Review your aggregated campaign summary from Task 3:

1. Which promotional code generated the highest total revenue?
2. What was the average order spend for customers who used no promotion compared to `PROMO-A`?
3. Draft a 2-3 sentence recommendation memo to Liam Vance evaluating campaign effectiveness.